Supplementary Material: Choosing a Fast Stand-In for a Slow Simulation

[Author names to be supplied by the authors]

S1. Proofs

Both results are standard; we include short proofs for completeness.

Proof of proposition 1main.pdf.. Let f^{-i} minimise $\sum_{j\neq i}(y_j - f_j)^2 + \lambda N \|f\|_K^2$ and let y' equal y except $y'_i = f_i^{-i}$. For any f, the objective with data y' equals the leave-one-out objective plus $(y'_i - f_i)^2 \geq 0$, so f^{-i}, which makes the extra term zero and minimises the first, also minimises the objective with data y'. By uniqueness of the ridge solution, $f^{-i} = Hy'$. Hence $f_i^{-i} = (Hy')_i = \hat{y}_i - H_{ii}y_i + H_{ii}f_i^{-i}$, so $(1-H_{ii})f_i^{-i} = \hat{y}_i - H_{ii}y_i$ and $y_i - f_i^{-i} = (y_i - \hat{y}_i)/(1-H_{ii})$. For $K = Q\,\mathrm{diag}(w)Q^\top$ with Q orthogonal, $H = Q\,\mathrm{diag}(w/(w+\lambda N))Q^\top$ and, since $\sum_j Q_{ij}^2 = 1$, $1 - H_{ii} = \sum_j Q_{ij}^2\, \lambda N/(w_j + \lambda N)$. For the polynomial model with a thin design matrix $F = U\,\mathrm{diag}(s)V^\top$, $1 - H_{ii} = (1 - \sum_j U_{ij}^2) + \sum_j U_{ij}^2 \lambda N/(s_j^2 + \lambda N)$. A bug in an early version of the tool (section S4, V9) came from computing $1 - H_{ii}$ by subtraction and clipping it. □

Proof of proposition 2main.pdf.. Let $P = V_r V_r^\top$. For $x \in \bar{\mathbf{u}} + \mathrm{span}(V_r)$, $x - \mathbf{u} = (x - \bar{\mathbf{u}} - P(\mathbf{u} - \bar{\mathbf{u}})) - (I - P)(\mathbf{u} - \bar{\mathbf{u}})$, where the first term lies in $\mathrm{span}(V_r)$ and the second in its orthogonal complement. By Pythagoras, $\|x - \mathbf{u}\|_2^2 \geq \|(I - P)(\mathbf{u} - \bar{\mathbf{u}})\|_2^2$. Dividing by $\|\mathbf{u}\|_2$ and averaging over runs gives the claim. □

S2. Pre-registration record

Hypotheses, success criteria and analysis code were committed to the project's git repository before the results they concern (a self-hosted history, not an independent registry). Table S1 lists, for each experiment, the protocol commit and the first commit that contains its results; the check is automatic (`audit/prereg_order_check.py`) and shows that no protocol was committed after its results. It shows protocol-before-results in history, not

protocol-before-first-run. Experiments H, G3 and the Stage-6 experiments T, C and S are labelled post hoc in motivation: they were designed in response to earlier results or to reviews (section S4).

Table S1: Order of pre-registration and results commits.

Exp	Protocol commit	First results commit	Hours before	Order ok
A	bcf7f30	2d0e2d4	9.33	yes
D	bcf7f30	2abe263	10.99	yes
B	bcf7f30	72cb02c	11.07	yes
A2	72cb02c	0bcb608	1.49	yes
E	72cb02c	703ad59	2.38	yes
A3	c9a7b93	82dbbe2	1.65	yes
G	c9a7b93	39b1b7f	3.66	yes
F	c9a7b93	39b1b7f	3.66	yes
H8	48a0698	84f2a21	0.08	yes
H32	48a0698	84f2a21	0.08	yes
G3	4cd411a	b37dd4e	0.05	yes

S3. Earlier-stage hypotheses and verdicts

Table S2 lists the hypotheses and primary tests of the earlier stages (original 16 problems; independent replication on eight new problems; mechanism controls; longer-training and precision checks). Most confirmatory tests of the first stage were not supported; the one supported at cell level (unequal tuning) rests on 5 of 16 problems.

Pilot diagnostic.. A frozen rule built from the original problems was tested on eight new problems from four new families: $\rho = -0.38$ (exact $p = 0.18$); out-of-sample gain 0.60 with Holm-adjusted $p = 0.051$ (10^6 bootstrap draws); the threshold rule classified 6 of 24 non-tie cells correctly. The pilot statistic also correlates with the number of inputs, and all problems with $f_1 < -2$ among the 24 have three inputs; we conclude that the diagnostic is inconclusive and do not use it in the tool.

Robustness.. Class labels agree between halves of the random-search draws in 83% of cells (sign agreement when both clear: 99%); 80% of reduced-basis-clear cells and 57% of network-clear cells have all three seeds beyond the tie band; class labels computed from the median instead of the mean error agree in 95% of cells; the extrapolation test set gives a median $\bar{y}$ of -0.03 against 0.16 on the interpolation set at $N = 1024$.

Table S2: Earlier-stage tests. $\bar{y}$ as in (3).

ID	Criterion	Observed	p	Verdict
H1a (v1)	median $\bar{y}$ at N=32 positive (reduced basis better)	median -0.059	0.874	not supported
H1b (v1)	$\bar{y}$ falls with N in most problems	5/16 problems	0.962	not supported
H2 (v1)	pilot features beat $\{\log_2 N, p\}$	CI [-0.07, 6.40]	0.145 (Holm)	not supported
H4a (v1)	unequal tuning moves wins toward the more tuned side	10 vs 1 discordant cells	0.006	supported (cell level)
P1 (Exp A)	Spearman($f_1, \bar{y}$)< 0, 8 new problems	$\rho = -0.38$	0.180	not rejected
P2 (Exp A)	frozen rule beats baseline out of sample	gain 0.60	0.051 (Holm)	not rejected (knife-edge)
Q1/Q2 (A2)	≥ 4 of 7 smooth candidates with $\bar{y} \geq 1$; none network-clear	7/7; 0	–	met (confounded with p)
D1	narrow-range kernel ridge within 3× of GP in $\geq 75\%$ of cells	median ratio 14.3; 13% within 3×	–	rejected
E	iso-GP vs ARD-GP and kernel ridge	ratios 1.92, 8.1	–	marginal-likelihood rule met (low p only)
G1	full-field GP within 3× of POD+GP	median ratio 0.95	–	supported
H (post hoc)	wide-range kernel ridge within 3× (B=8 or 32)	medians 6.5, 2.7	–	neither rule met (mixed)
B1/B2	4× epochs: sign kept in $\geq 85\%$; median shift < 0.5	88%; 0.24	–	met, but 15 of 48 classes change
F	float64 + L-BFGS: ≥ 5 of 7 decisive gaps remain ≥ 1 decade	5 of 7	–	met

S4. Deviations from the plan

Every change to the plan is recorded with its date and reason in the repository (`research_design/deviations.md` for the first stage and `deviations_v2.md` afterwards); the text is reproduced here.

Deviations, disclosures and post-hoc (exploratory) analyses

*	*	*	*	*	
0.2000	0.2000	0.2000	0.2000	0.2000	Effect on confirmatory results
*	*	*	*	*	
0.2000	0.2000	0.2000	0.2000	0.2000	Effect on confirmatory results
*	*	*	*	*	
0.2000	0.2000	0.2000	0.2000	0.2000	restarts	none	launch attempts	5	killed (BLAS thread oversubscription) before any job finished; the container restarted twice later and killed in-flight jobs.

() () () () ()
* * * * *

0.20.20.40.20.0Type Detail Effect on confirmatory results 0.2000

() () () () ()
* * * * *

0.20.20.20.20.20.0 none

Interim analysis | closure | A premature analysis was run when 47 of 48 jobs existed (seeds 0,1 complete; my file count wrongly included `environment.json`). It used only complete seed batches (0,1) and was

*	*	*	*	*	
0.2000	0.2000	0.2000	0.2000	0.2000	Effect on confirmatory results
*	*	*	*	*	
0.2000	0.2000	0.2000	0.2000	0.2000	Held-out seeds

to 0–2 (mandatory core); extension seeds 3–4 **not run** (compute/time).

Type	Detail	Effect on confirmatory results
Post-hoc exploratory analyses	Exploratory after seeing confirmatory results; every one is labelled *exploratory* in the manuscript and none changes a confirmatory verdict. See `results/exploratory/`.	None

*	*	*	*	*		
0.2	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	Effect detail on confirmatory results
					0.2000	
*	*	*	*	*		
0.2	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	Could not reproduce search space and re-sample fixed (bug) hyperparameters; did not seed the RNG; Python's per-process-salted `hash(method)`, so trials 1-7 of every hyperparameter search are sampled differently in
					9	

Deviations, disclosures and corrections in the v2 / v2.1 / v2.2 cycle

() () () () ()
* * * * *

0.2000 0.2000 0.2000 0.2000 Type Detail Effect on pre-registered verdicts

0.2000

() () () () ()
* * * * *

0.2000 0.2000 0.2000 0.2000 Type Detail Effect on pre-registered verdicts

0.2000

() () () () ()
* * * * *

0.2000 0.2000 0.2000 0.2000 0.2000

B shift sign convention

found by independent audit (F1), fixed **before Exp B was analysed**

= y6000 − y1500 is negative when NL improves; B2 "shift toward NL < 0.5" would have passed trivially. Replaced by `shift_toward_NL = y1500`

(B not yet analysed)

	()	()	()	()	()
*	*	*	*	*	
0.2	0.0	0.0	0.0	0.0	Effect on pre-registered ver-
				0.2000	dicts

	()	()	()	()	()
*	*	*	*	*	
0.2	0.0	0.0	0.0	0.0	none

baseline-2/20; p0 or coverage in recount) counterdemonstrators_L1v2.fi found by in-degree-per-dem audited, verified before **A2 ran**: p0 = 0.15, success k 4 of 7. 13 tual (A2 ter- not yet run) and 3/20 (burg- ers_L1 fi- d 1.9972, ȳ = 2.25). Amended in ad- den- dum v2.2

()	()	()	()	()		
*	*	*	*	*		
0.2000	0.2000	#0.2000			Type	Detail	Effect on pre-registered verdicts

0.2000	0.2000	0.2000	0.2000	0.2000	Type	Detail	Effect on pre-registered verdicts
() *	() *	() *	() *	() *	Not agnostic verdict wording	characterisation ("-cated" → *pre-registered criteria not met; inconclusive; carried by one family (nl-heat 88.3 % of P2 gain; = +0.49 without it); confounded with p (Spear-man*	None (characterisation label's unchanged)

() () () () ()
* * * * *

0.20.20.40.2000 Type Detail Effect on pre-registered verdicts

0.2000

() () () () ()
* * * * *

0.20.20.20.20.2000 Stage Outcome

Chronology | Disclosure | Stage 3 (addendum v2.1) was committed within minutes after the Exp A and Exp D results; Stage 4 (addendum v2.2) after two independent audits. All

*	*	*	*	*
0.2	[illegible]	[illegible]	[illegible]	Effect on pre-registered ver-
			0.2000	dicts

*	*	*	*	*
0.2	[illegible]	[illegible]	[illegible]	none

search drawn from a pre-registered, founder-only conditioning” (F2 seed 1 method): 16 distinct sets per method in the whole study)

RNG changes by only 1 bit

(“adjust the sensitivity figure used for every instance and within a seed (3 distinct sets per method in the whole study)

sensitivity analysis planned in advance

described; 7 simulated datasets

*	*	*	*	*		
0.2000	0.2000	0.2000	0.2000	0.2000	type	Effect on pre-registered ver-dicts
					0.2000	dicts

*	*	*	*	*		
0.2000	0.2000	0.2000	0.2000	0.2000	0.2000	one
ral sig =					on	
batch	mod	Ste-				
count		ent	for	dicts		
	found	N	d			
	by	512				
	au-	(1500				
	dits	steps)				
	(F02)		02)	t		
			512			
			for			
			N			
			=			
			1024			
			(3000			
			steps):			
			the			
			neu-			
			ral			
			step			
			count			
			dou-			
			bles			
			at			
			N			
			=			
			1024.			
			Exp			
			F			
			re-			
			moves			
	17		this			
			(full			
			batch).			

() () () () ()
* * * * *

0.20.20.20.20.20#IDDateTypeDetail Effect on pre-registered verdicts

0.2000

() () () () ()
* * * * *

0.20.20.20.20.20 none
fixes keeping test count fixed (36 instances); weaker leakage tests replaced; BLAS thread env set at the top of entry scripts.

V9 — SurroSelect smoke-test bug (before any registered Stage-6 run) — 2026-11-02

During a smoke test of the tool on the new problem `lshape_L1` (seed 0, N = 64 and 256) and a quick check of the KRR and polynomial members on

`heatcomp_L2` and `convdiff_L1` (N = 64, 256), the wide-range kernel-ridge member returned a test error of 0.5 (10x worse than any other member). Cause: the closed-form leave-one-out residual divides by 1 - H_ii, which was clipped at 1e-6 (and computed by cancellation), producing spuriously tiny LOO scores for near-interpolating fits; the length-scale grid was also defined for inputs on [0,1] but with the [-1,1] limits of the benchmark's wide-range control. Fix (committed before any registered run): 1 - H_ii computed without cancellation (KRR: sum_j Q_ij^2 lam N/(w_j + lam N); polynomial: (1 - sum U_ij^2) + sum U_ij^2 lam N/(s_j^2 + lam N)), no clipping above 1e-12, ridge values with condition number above 1e12 skipped, length-scale grid [0.05, 15] · sqrt(p) on [0,1]-scaled inputs (= [0.1, 30] · sqrt(p) on [-1,1]); a regression test was added. The smoke tests showed test errors of all pool members on lshape_L1 (seed 0, N = 64, 256) and of the KRR and polynomial members on three problems; the pool, criteria, tie band, folds and all pre-registered thresholds were NOT changed after seeing them. Addendum v3.0 stated that no surrogate error on the new problems had been seen; this entry corrects that statement.

S5. Data-leakage audit

Training, test and extrapolation inputs are drawn from separate named random streams. Over all 44 problems the number of exact input matches between the training pool and the test or extrapolation sets is 0. A planted-leak test (20 test samples copied into the training set of problem `heatcomp_L1`, $N = 64$) is flagged by the overlap check (20 exact matches, none without the leak), and the error on the leaked samples is 229 times lower than on the other test samples, showing that a leak would be visible. Validation errors do not depend on test labels (unit test, corrupting all test labels). Fitting steps (POD, scalers, hyper-parameters, cross-validation folds) use training runs only. Two properties remain: training subsets are nested within a seed, and seeds share the pool and test set, so seed variance understates sampling variance at large N. Comparisons in the mechanism study that use the best network by *test* error are oracle comparisons and are labelled as such.

S6. Per-problem results

Table S5 gives $\bar{y}$ (reduced basis against direct networks, validation-selected members, seed means) for every problem; positive values favour the reduced basis.

Table S5: $\bar{y}$ per problem and budget (positive: reduced basis better).

Problem	p	N=64	N=256	N=1024
advdiff_L1	4	-0.04	0.17	0.21
advdiff_L2	4	-0.09	-0.00	-0.07
bratu_L1	3	2.78	2.98	3.31
bratu_L2	3	2.14	2.97	3.08
burgers_L1	4	-0.06	0.68	2.25
burgers_L2	4	0.01	0.08	0.11
convdiff_L1	3	-0.03	0.60	0.48
convdiff_L2	4	-0.05	0.10	0.04
darcy_L1	4	0.15	1.12	1.13
darcy_L2	10	-0.00	-0.05	-0.19
darcyhd_L1	16	0.04	-0.02	-0.13
darcyhd_L2	32	0.03	0.00	-0.07
darcylin_L1	16	0.18	0.63	0.48
darcylin_L2	32	0.04	0.43	0.18
diff1d_L1	4	0.89	2.36	2.70
diff1d_L2	8	0.17	0.62	1.00
fisher_L1	4	-0.21	0.09	0.52
fisher_L2	4	-0.26	-0.26	-0.44
heat2d_L1	4	0.18	0.93	1.38
heat2d_L2	4	-0.04	-0.10	-0.34
heatcomp_L1	4	0.00	0.52	0.88
heatcomp_L2	9	-0.01	-0.06	-0.12
heatmove_L1	3	0.13	0.13	-0.17
heatmove_L2	5	-0.04	-0.02	-0.15
heatmv3_L1	3	0.12	-0.11	-0.44
heatp_L1	3	0.77	2.49	2.87
heatp_L2	3	1.08	2.71	2.90
helm_L1	3	1.20	2.69	2.93
helm_L2	3	-0.16	2.08	2.64
incl3_L1	3	0.02	0.05	-0.21
incl_L1	4	-0.02	-0.00	-0.25
incl_L2	4	0.02	-0.01	-0.13
kdv_L1	4	-0.14	-0.26	-0.23
kdv_L2	4	-0.04	-0.03	0.29
lshape_L1	4	0.20	1.14	1.66
lshape_L2	7	-0.11	0.01	-0.01
nlheat_L1	3	1.25	2.51	2.72
nlheat_L2	3	0.75	2.16	2.58
reacdiff_L1	3	0.85	1.62	2.57
reacdiff_L2	3	0.57	1.96	2.38
transport_L1	4	-0.26	0.05	0.75
transport_L2	4	-0.32	-0.23	-0.33
wave_L1	4	0.16	0.29	0.78
wave_L2	4	0.10	-0.06	-0.04

S7. The tool

Installation and use.. The package `surroselect` (directory `surroselect/` of the repository; MIT licence, copyright holder to be set by the authors) depends on NumPy, SciPy and scikit-learn; PyTorch and Optuna are needed only for the optional `TunedMLP`. In Python:

```
sel = SurrogateSelector(cv=4).fit(X, Y)    # X: runs x inputs, Y: runs x outputs
sel.learning_curve()
print(make_report(sel, target=0.01))
Y_new = sel.predict(X_new)
```

On the command line, with plain `.npz` files that hold arrays `X` and `Y`:

```
surroselect fit train.npz --model m.pkl --target 0.01 --report report.md
surroselect predict m.pkl X_new.npy --out Y_pred.npy
surroselect evaluate m.pkl test.npz
```

Default pool.. POD+GP (Matérn-5/2, rank 32), POD+GP (squared exponential, rank 32), POD+GP (Matérn-5/2, rank 8), POD+KRR with a wide grid of kernel, length-scale and ridge selected by closed-form leave-one-out error, POD+polynomial (degree up to six, at most 3000 features, leave-one-out), and a GP on the full field (Matérn-5/2 or squared exponential; only for $m \leq 2000$). An optional metric, the normalised RMSE $\|Y - \hat{Y}\|_F / \|Y - \bar{Y}_{\text{train}}\|_F$, is available for outputs with a large offset.

Validation of the tool on plain files.. To check the tool itself, without the benchmark pipeline in the loop, three of the new problems (`heatcomp_L1`, `lshape_L1`, `convdiff_L1`; $N = 256$) were exported to `.npz` files and processed with the three command-line commands as separate processes. All of sequences ran without error, and the error printed by `evaluate` agreed with an independent computation from the saved predictions to within 10^{-12} in of cases (`results/s6/plain_files_validation.json`).

Tests.. 9 unit tests for the tool (closed-form leave-one-out against brute force and its stability near interpolation, exact polynomial recovery, selection accuracy, the basis-limit flag, the power-law inversion, the report, the metric option and the command-line round trip), 6 tests for the finite-element problems and the 56 tests of the benchmark package are collected by the repository; `results/logs/tests_*.log` records the final runs. A fitted selector could not be saved in an early version after the metric option was added (deviation V10); the plain-file validation found it.

Known limits.. The tool assumes exchangeable runs (A3); it does not model noise; for $p \gtrsim 10$ the default GP kernels have many hyper-parameters fitted on at most 300 runs; the forecast is an extrapolation without guarantee; and the optional neural member uses a small search budget.